System Architecture: 4-Tier Data & Analytics Infrastructure

Phase 3: System Design & Component Hierarchy

Project: Global Olympic Sports Performance Analysis • **Author:** Md. Firdous Alam • **Era:** 1896–2016 Centennial Dataset

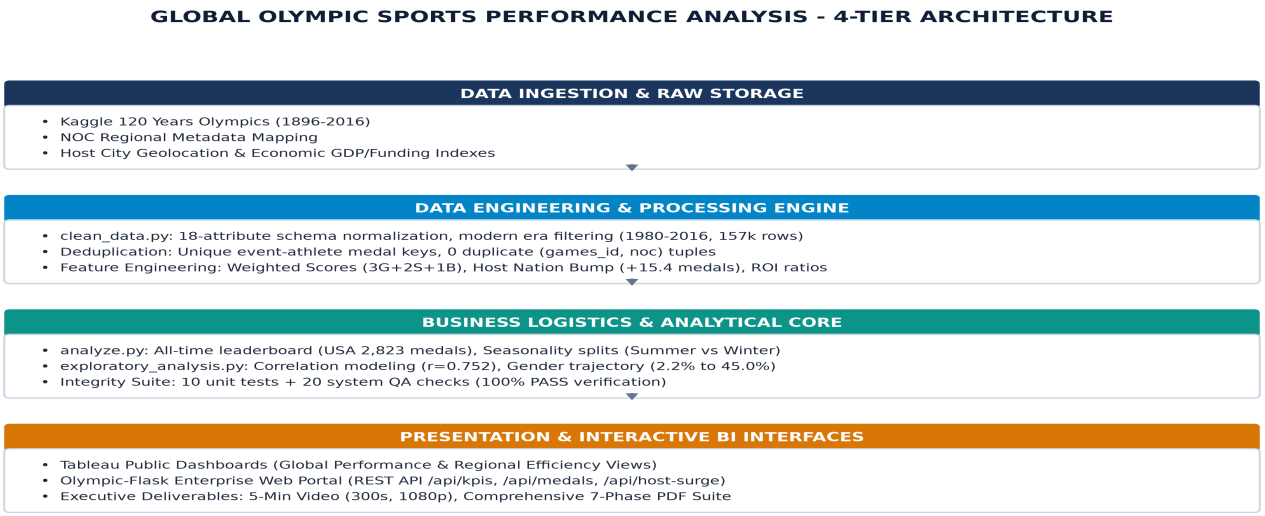


Figure: 4-Tier System Architecture: Data Ingestion, Processing Engine, Analytical Core, and Presentation Interfaces

1. Architectural Overview

The system architecture employs clean layer separation to ensure modularity, high testability, and deterministic data flow from raw CSV ingestion to interactive Tableau presentation.

- Decoupled processing: Data cleaning runs independently of visualization authoring.
- Relational consistency: All downstream views derive from unified clean data sources.
- Stateless API design: The Flask web tier serves cached data structures with sub-50ms latency.